

Problem 1

By Lagrange's Theorem we know that S_4 has subgroups of order 1, 2, 3, 4, 6, 8, 12

Let H be a subgroup of order 12. By Cauchy's Theorem we know it has a subgroup of order 3. Thus, it has a 3 cycle. Since it has a 3 cycle we know it is A_4 since A_4 is generated by 3 cycles, and this is normal since it has index 2.

Let H be a group of order 8. These must be 2-Sylow subgroups and by Sylow 3 we know there is 1 or 3 but there is 3 since two examples are $\langle (1234), (13) \rangle$ and $\langle (1243), (14) \rangle$ Thus these are all conjugate by Sylow 2 thus these are not normal.

Let H be a group of order 6. This must be S_3 since there is no elements of order 6 in S_4 . This means it has a 3 cycle but since all 3 cycles are conjugate we know that we can get an element outside the subgroup since the S_3 subgroup will only have 2 3-cycles.

Let H be a group of order 4. This is either C_4 or $C_2 \times C_2$. If it is C_4 then it is a cycle group generated by a 4 cycle, and since $\sigma(abcd)\sigma^{-1} = (\sigma(a)\sigma(b)\sigma(c)\sigma(d))$ we know that all 4 cycles are conjugate. Thus, since there 6 4-cycles we know we can get outside our group of 4.

If it is $C_2 \times C_2$ then if it has a transposition then we can get out again by similar proof by showing all transpositions are conjugate. Thus the only possibility is $\langle (12)(34), (13)(24) \rangle$

Since these are a product of disjoint cycles if we conjugate $\sigma(ab)(cd)\sigma^{-1} = (\sigma(a)\sigma(b))(\sigma(c)\sigma(d))$ then since σ permutes elements we know that this will also result in a product of disjoint transpositions and we have them all in this subgroup this is normal.

Let H be a group of order 3. Similar to order 4 for C_4 this is not normal

Let H be a group of order 2. Similar to order 4 for C_4 this is not normal

Thus we only have normal groups $\langle (12)(34), (13)(24) \rangle$, A_4 and $\{e\}$

For normal in A_4 we know from previous homework that it has normal $C_2 \times C_2$ and this could only be $\langle (12)(34), (13)(24) \rangle$

Thus all composition series are

$$S_4 \supseteq A_4 \supseteq \langle (12)(34), (13)(24) \rangle \supseteq \{1, (12)(34)\} \supseteq \{e\}$$

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Problem 2

- (a) For A_4 , Consider the homomorphism $\varphi : A_4 \rightarrow A_4 / \langle (12)(34), (13)(24) \rangle$ since $|\langle (12)(34), (13)(24) \rangle| = 4$ and is normal we know $A_4 / \langle (12)(34), (13)(24) \rangle \cong C_3$ which is abelian.
 Consider $[a, b] \in A'_4$, $aba^{-1}b^{-1}$. $\varphi(aba^{-1}b^{-1}) = \varphi(a)\varphi(b)\varphi(a)^{-1}\varphi(b)^{-1} = e$ since it is abelian thus $A'_4 \in \ker(A_4 / \langle (12)(34), (13)(24) \rangle)$ and the kernel of this is $\langle (12)(34), (13)(24) \rangle$ thus $A'_4 \leq \langle (12)(34), (13)(24) \rangle$
 Consider $(123)(12)(34)(132)(12)(34) = (13)(24)$, $(124)(12)(34)(142)(12)(34) = (14)(23)$, and $(13)(24)(14)(23) = (12)(34)$ thus we know $A'_4 = \langle (12)(34), (13)(24) \rangle$ and since $\langle (12)(34), (13)(24) \rangle$ is abelian $\langle (12)(34), (13)(24) \rangle' = \{e\}$
 Thus it is solvable and the derived series is $A_4 \supseteq \langle (12)(34), (13)(24) \rangle \supseteq \{e\}$
 For C_{12} since it is abelian its derived series is $C_{12} \supseteq \{e\}$ thus it is solvable
 For D_6 we know it has a normal subgroup $\langle x^2 \rangle$ thus we know $D_6 / \langle x^2 \rangle$ is a group of order 4 which we know is abelian thus from similar proof to above we know $D'_6 \leq \langle x^2 \rangle$
 Consider $yx^2y^{-1}x^4 = x^8 = x^2$ thus $D'_6 = \langle x^2 \rangle$ and since $\langle x^2 \rangle$ is abelian its commutator subgroup is $\{e\}$ thus it is solvable
 $D_6 \supseteq \langle x^2 \rangle \supseteq \{e\}$
 For $C_6 \times C_2$ it is abelian thus $C_6 \times C_2 \supseteq \{e\}$ and it is solvable
 For $C_3 \rtimes C_4$ we know it has a normal subgroup C_3 thus $(C_3 \rtimes C_4) \setminus C_3$ is order 4 and thus abelian thus $(C_3 \rtimes C_4)' \leq C_3$
 Consider $bab^{-1}a^2 = a^4 = a$ thus $C_3 = (C_3 \rtimes C_4)'$ and since C_3 is abelian, $C'_3 = \{e\}$
 Thus it is solvable with $C_3 \rtimes C_4 \supseteq C_3 \supseteq \{e\}$
- (b)
- For $n = 1$, $S_1 = \{e\}$ thus done, and it is solvable
 - For $n = 2$, $S_2 \cong C_2$ thus it is abelian thus $S_2 \supseteq \{e\}$, and thus solvable
 - For $n = 3$, $S_3 \cong D_3$ thus it has a normal subgroup C_3 and since $D_3 / C_3 \cong C_2$ it is abelian thus $D'_3 \leq C_3$
 Consider $xyx^{-1}x^2 = x^4 = x$ thus $C_3 \leq D'_3$ thus $D'_3 = C_3$ thus since it is abelian $C'_3 = \{e\}$ thus it is solvable with $S_3 = D_3 \supseteq C_3 \supseteq \{e\}$
 - For $n = 4$, we know S_4 has normal subgroup A_4 thus $S_4 / A_4 \cong C_2$ thus abelian thus $S'_4 \leq A_4$
 Consider $(14)(123)(14)(132) = (142)$ thus since S'_4 is normal we can conjugate this 3-Cycle to get all 3-Cycles and thus $S'_4 = A_4$ and for A_4 look above thus it is solvable
 $S_4 \supseteq A_4 \supseteq \dots$
 - For $n \geq 5$ by similar proof above we can show that $S'_n = A_n$ and since A_n is simple we know that $A'_n = \{e\}$ or A_n if it is $\{e\}$ this would mean it is abelian which is not true thus it is A_n thus the derived series is $S_n \supseteq A_n \supseteq A_n \dots$ thus not solvable

Problem 3

	y	y		x	x	x	$\dots$	x	x		x	y	x	y	
1	1		1	2	3	4	$\dots$		n	1	2	n		1	1
2	n		2	3	4	5	$\dots$		1	2	3	$n-1$		n	2
3	$n-1$		3	4	5	6	$\dots$		2	3	2	$n-2$		$n-1$	3
$\vdots$	$\vdots$		$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\dots$		$\vdots$	$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$
n	2		n	1	2	3	$\dots$		$n-1$	n	1	1		2	n

Thus we know since $1 = \{y\}$ is index n and does not act trivially we know G has order $2n$

Problem 4

$$x^2 = yxy$$

$$x^2y^2 = yx$$

$$x^4 = e = yxyxyx$$

$$e = x^2y^3xy$$

$$e = x^3y$$

$$x = y$$

Thus the order of x and y both divide 3,4 thus they have order 1 thus $x = y = e$

	y	y	y		x	x	x	x		x	x	y	x	y	
1	1	1	1	1	2	3	4	1	2	3	4	1	1	2	1
2	3	4		2	3	4	1	2	3	4	2	4		2	2
3	4	2		3	4	1	2	3	4	1	1	2		3	3
4	2	3		4	1	2	3	4	1	2	3	3		4	4

We can see from the table that from the yellow $2 = 3$, from the blue $4 = 2$

	y	y	y		x	x	x	x		x	x	y	x	y	
1	1	1	1	1	2	2	2	1	2	2	2	1	1	2	1
2	2	2		2	2	2	1	2	2	2	2	2		2	2

We can see from the green that $2 = 1$ thus this is a trivial table thus $x = y = e$

Problem 5

(a)

	z z z z z					x x		y y y			x y z		
1	1	1	1	1	1	1	2	1	6	2	1	2	1
2	3	4	5	6	2	2	1	2	1	6	2	1	6
3	4	5	6	2	3	3	6	3	5	7	3	6	2
4	5	6	2	3	4	4	7	4	11	8	4	7	3
5	6	2	3	4	5	5	8	5	7	3	5	8	4
6	2	3	4	5	6	6	3	6	2	1	6	3	5
7	8	9	10	11	7	7	4	7	3	5	7	4	11
8	9	10	11	7	8	8	5	8	4	11	8	5	7
9	10	11	7	8	9	9	11	9	10	12	9	11	8
10	11	7	8	9	10	10	12	10	12	9	10	12	9
11	7	8	9	10	11	11	9	11	8	4	11	9	10
12	12	12	12	12	12	12	10	12	9	10	12	10	12

Since z does not act trivially we know that $[G : \langle z \rangle] = 12$ thus $|G| = 60$

- (b) From the table above since all of the elements do not act trivially we know since they divide a prime order they must be that prime order so z is order 5, x is order 2, and y is order 3.

Consider $z = y^2x$ thus we can rewrite G to be $G = \langle x, y \mid x^2 = y^3 = (y^2x)^5 = e \rangle$

Let $\varphi : G \rightarrow A_5$ s.t. $\varphi(x) = (12)(34)$, and $\varphi(y) = (235)$

Homomorphism:

We only have to check $(\varphi(y)^2\varphi(x))^5 = e$

$((253)(12)(34))^5 = e$ thus this is a homomorphism

Injective: Let $a \in \ker(\varphi)$

If $\varphi(a) = \varphi(x^i y^j) = \varphi(x)^i \varphi(y)^j = ((12)(34))^i ((235))^j = e$

If i is even then $= (234)^j = e$ thus j is a multiple of 3 thus $a = e$

If j is odd then $= (12)(34)(235)^j = e$

If j is a multiple of 3 then $(12)(34) = e$ which is not possible so either $= (12)(34)(235)$ or $= (12)(34)(253)$

If the first case then $(12)(34)(235) = e$ this is not possible

If the second case then $(12)(34)(253) = e$ this is not possible

If $\varphi(a) = \varphi(y^j x^i) = \varphi(y)^j \varphi(x)^i = (235)^j ((12)(34))^i = e$

Similar proof as above. Thus, $a = e$

Thus injective

Thus since $|G| = |A_5|$ and φ is injective φ is bijective thus $G \cong A_5$

Thus consider H use φ , and we get that $\varphi(H) \leq A_5$ call this subgroup K

Consider K , it is generated by $\varphi(y^2xyxy) = (142)$ and $\varphi(x) = (12)(34)$

Since K is generated by even permutations with only 4 elements we know that $K \leq A_4$

We know K has at least 5 elements $\{e, (142), (124), (12)(34), (142)(12)(34) = (243)\}$ thus $|K| = 12$ since if $|K| = 6$ then since G has no element of order 6 it must be S_3 and S_3 has 3 elements of order 2 thus K must have 3 double transpositions but these form a subgroup of order 4 but 4 does not divide 6 thus $|K| = 12$ Thus G acts on G/H by left multiplying on H 's $[G : H] = 5$ cosets thus $\phi : G \rightarrow S_5$

$$H \cong K \cong A_4$$

(c) I did that above

(d) Consider K use φ , and we get that $\varphi(K) \leq A_5$ call this subgroup B

Consider B , it is generated by $\varphi(yxy^2x) = (15234)$ and $\varphi(x) = (12)(34)$

(15234) is order 5, and $(12)(34)$ is order 2.

Consider $(12)(34)(15234)(12)(34) = (14325) = (15234)^{-1}$ thus $B \cong D_5$

Thus $K \cong B \cong D_5$ thus $|K| = 10$

Thus G acts on G/K by left multiplying on K 's $[G : K] = 6$ cosets thus $\phi : G \rightarrow S_6$

Problem 6

(a) Consider $(xyx^{-1}y^{-1}R) = R$ this is true since $[x, y] \in R$

Since R is normal $xRyR = yRxR$

Thus $x^2Ry^2R = xRyRxRyR = y^2Rx^2R \implies [x^2, y^2]R = R$ thus $[x^2, y^2] \in R$

(b) Consider $c = [x, y]$ this is an element of F' by definition.

Since F' is normal it also has all conjugates of c , thus $R \subseteq F'$ since R is the smallest normal group with c .

Shown in (a) since the two generators of F commute in F/R we know that F/R is abelian.

Thus shown in problem 2 we know that $F' \leq R$ thus $F' = R$